Ashwath E

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.75
Maharishi Vidya Mandir	Class XII (CBSE)	2023	95.20%
The Schram Academy	Class X (CBSE)	2021	97.80%

SCHOLASTIC ACHIEVEMENTS

Contributor to SciPy 1.18.0 - fixed bugs in root-finding algorithms and wrote unit tests.

Won 2nd place in Inter-IIT Tech Meet 13.0 in Pathway's problem statement by designing a dynamic multi-agentic RAG system capable of reasoning, risk analysis, adaptive retrieval, statistical summarization and code writing and execution. (Fully descriptive blog with video demonstrations and explanations can be found here)

RESEARCH

BP-NMS: Linear-Time Bit-Parallel Non-Maximum Suppression for Scalable Object Detection -

- Devised an alternative approach to NMS which avoids FLOps entirely, bypassing its greatest compute bottleneck.
- Reduces time-complexity to from quadratic to linear time, significantly boosting throughput at little cost.

Discovery and Recovery of Typographic Information From Vision Models -

- Probed various vision model encoders to establish the existence of typographic features using different methods.
- Recovered the formatting information lost in decoding with intricately constructed fine-tuning objectives.

Physically-Informed MissForests for Streamflow Discharge Prediction -

- Tackled the problem of streamflow discharge prediction by devising intelligent, informed missforests.
- Engineered features from spatio-temporal data to accurately capture climatological trends.

WORK EXPERIENCE

AI Research Intern, Thomson Reuters - Worked on building state-of-the-art legal document parsing systems, evaluation methods and built legal deep research systems. Worked on investigating the retention of typographic signal in OCR models and vision encoders. Adapted the HRM architecture for language modeling and created small models from scratch.

Lead of R&D, semXai - Worked as a lead researcher to devise hardware-aware methods to compress LLMs, VLMs, world models and CNN-based vision models. Devised new algorithms which compete with state-of-the-art model pruning methods while simultaneously eliminating hardware latency and energy bottlenecks.

ML Engineer, Wanderlust - Working as an ML engineer in the product team at Japanese startup Wanderlust, writing AI-powered algorithms for intelligent document structure-aware parsing and transformation (ETL). Developing efficient, cutting-edge document transformation systems to enable easy creation of data for RAG systems and GenAI.

PROJECTS

Configurable RISC-V Simulator -

Wrote a configurable, terminal-based RISC-V simulator in C, which can generate machine code for assembly, simulate execution (with support for breakpoints, line-by-line execution, etc), memory and registers, with a fully customizable cache and cache statistic calculation.

VectorDB -

Designed and a high-performance C++ vector database with Python bindings using pybind11. Supports efficient storage, management, and similarity search of high-dimensional vectors. Provides both C++ and Python APIs with easy integration into machine learning workflows. Comes with an automated build system with CMake.

POSITIONS OF RESPONSIBILITY

Deputy Contingent Leader, Inter IIT Tech Meet 14.0

Overall Head at Epoch, the AI/ML & Data Science Club of IITH

Techy Domain Coordinator at Elan & nVision, IITH

Core at Epoch, the AI/ML & Data Science Club of IITH

SKILLS

Languages - Python, C, C++, JavaScript, HTML/CSS, SQL, Dart, Java, C#.

Frameworks - PyTorch, TensorFlow, Transformers, Pinecone, Weaviate, ChromaDB, Langchain, Langgraph, LlamaIndex, NLTK, Scikit-learn, Numpy, Pandas, Matplotlib, Seaborn, OpenCV, Flutter, NextJS, ReactJS, Express.js, Node.js, Prisma.

Technologies - Docker, Nginx, Firebase, Azure, GitHub/Git, PostgreSQL, MySQL, MongoDB, Google Apps Script, AWS.

Technical Skills - Full-stack web development, app development, game development, AI/ML solution & product development, cloud deployment and load balancing.

RELEVANT COURSES

Probability and Random Variables, Programming for AI, Matrix Theory, Linear Systems & Signal Processing, Foundations of Machine Learning, Applied Statistics, DSA, Algorithms, DBMS (1 & 2), OS (1), Compilers (1), Computer Architecture, Deep Learning, NLP, CV, RL, AI Mini Project (1 & 2), Fraud Analytics, Predictive Modeling, Marine Robotics, Control Systems.

EXTRACURRICULAR

Badminton - regional cluster tournaments and inter-school events

Football - regional cluster tournament runner-up